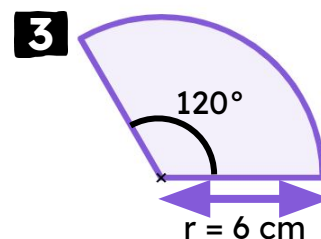
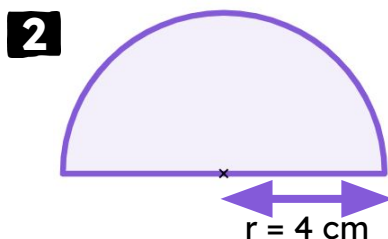
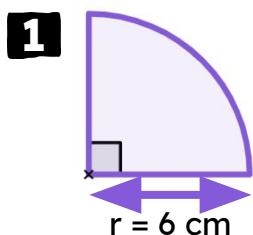




Area of composite shapes

Task A: Finding the area of sectors of a circle

1) Match the circle sectors with the possible correct calculations and answers. Some calculations do not match with a sector.



a $\frac{1}{2} \times \pi \times 2^2$

b $\frac{1}{3} \times \pi \times 6^2$

e 2π

f 8π

c $\frac{1}{2} \times \pi \times 4^2$

d $\frac{1}{4} \times \pi \times 6^2$

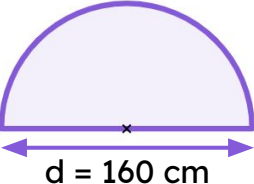
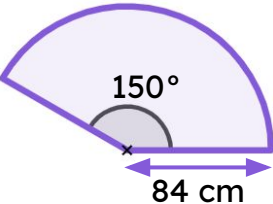
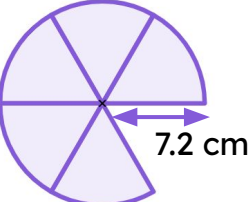
g 9π

h 12π

2) Complete the table. Give your answer both as a decimal (rounded to 1 d.p.) and in terms of π .

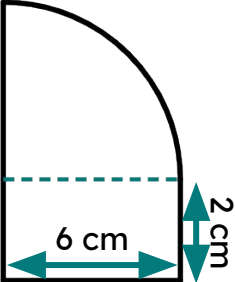
sector	step 1	step 2	answer (cm ²)
a)	$\frac{1}{2} \times \pi \times 30^2$	$\frac{1}{2} \times \pi \times 900$	
b)	$\frac{3}{4} \times \pi \times 18^2$		763.4
c)	$\text{---} \times \pi \times 57^2$		2166π

2) Complete the table. Give your answer both as a decimal (rounded to 1 d.p.) and in terms of π .

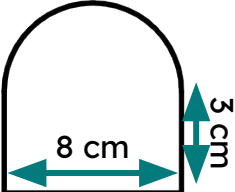
sector	step 1	step 2	answer (cm ²)
d) 			
e) 			
f) 			

Task B: Finding the area of composite shapes with sectors

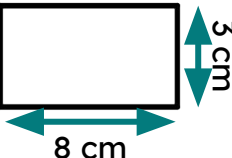
1) Match the composite shape to its component parts and calculations.



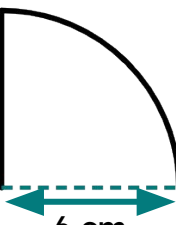
shape 1

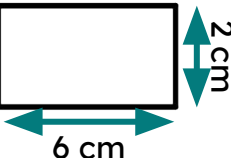


shape 2

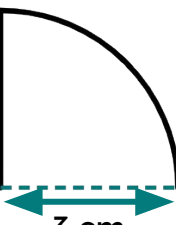
a 

b 

c 

d 

e 

f 

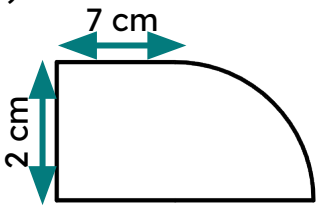
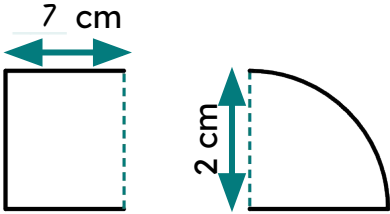
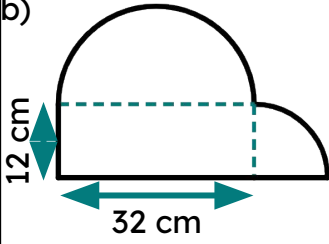
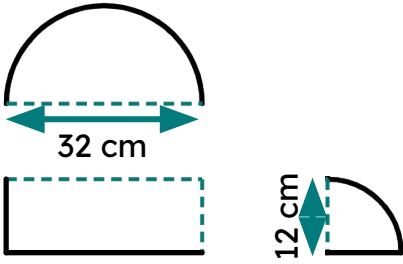
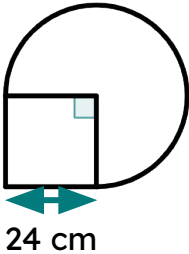
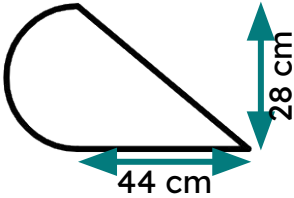
g $\frac{1}{2} \times \pi \times 4^2 + 24$

h $\frac{1}{2} \times \pi \times 8^2 + 24$

i $\frac{1}{4} \times \pi \times 6^2 + 12$

j $\frac{1}{2} \times \pi \times 3^2 + 12$

2) Complete the table. Give your answer as both a decimal and in terms of π

shape	shape broken into parts	calculations (addition)	answer
a) 		$2 \times \underline{\hspace{1cm}} + \frac{1}{4} \times \pi \times 2^2$	
b) 		$\frac{1}{4} \times \pi \times \underline{\hspace{1cm}}^2 +$ $\frac{1}{2} \times \pi \times \underline{\hspace{1cm}}^2 +$ $\underline{\hspace{1cm}} \times \underline{\hspace{1cm}}$	889.2 cm ²
c) 			
d) 			

3) What do you notice about the areas of each shaded region?

Show your working to support your observations.

